

UNIT II

Speech Recognition

Content

- Real and Complex Cepstrum
- Application of cepstral analysis to speech signal
- Feature extraction for speech
- MFCC
- LPCC
- Static and dynamic feature for speech recognition
- Distance measures
- Vector quantization models
- HMM
- Robustness issues
- Discrimination in the feature space
- Feature selection
- Gaussian Mixture model

Content

- Feature extraction for speech
 - Features
 - Feature Extraction
 - Type of features

Features

- Feature is a numeric or categorical representation of raw data that isolates a specific, meaningful attribute for analysis.
- Features may be considered as the smallest, fundamental building blocks of speech sounds.
- Features describe the psychological and physiological components that make up a single sound (phoneme).
- A speech feature is a measurable data point extracted from a raw audio signal.
- Features can be of various types based on different aspects.

Need for Features Extraction

- Raw data is often too large, noisy, or complex for algorithms to process effectively.

- Features are extracted to:

1. Reduce Dimensionality:

Turn complex data into a concise set of descriptive variables.

2. Remove Noise:

Ignore irrelevant information (like background static in an audio file) and keep only what matters.

3. Enable Pattern Recognition:

Provide clean inputs so machine learning models can find trends, classify items, or make accurate predictions.

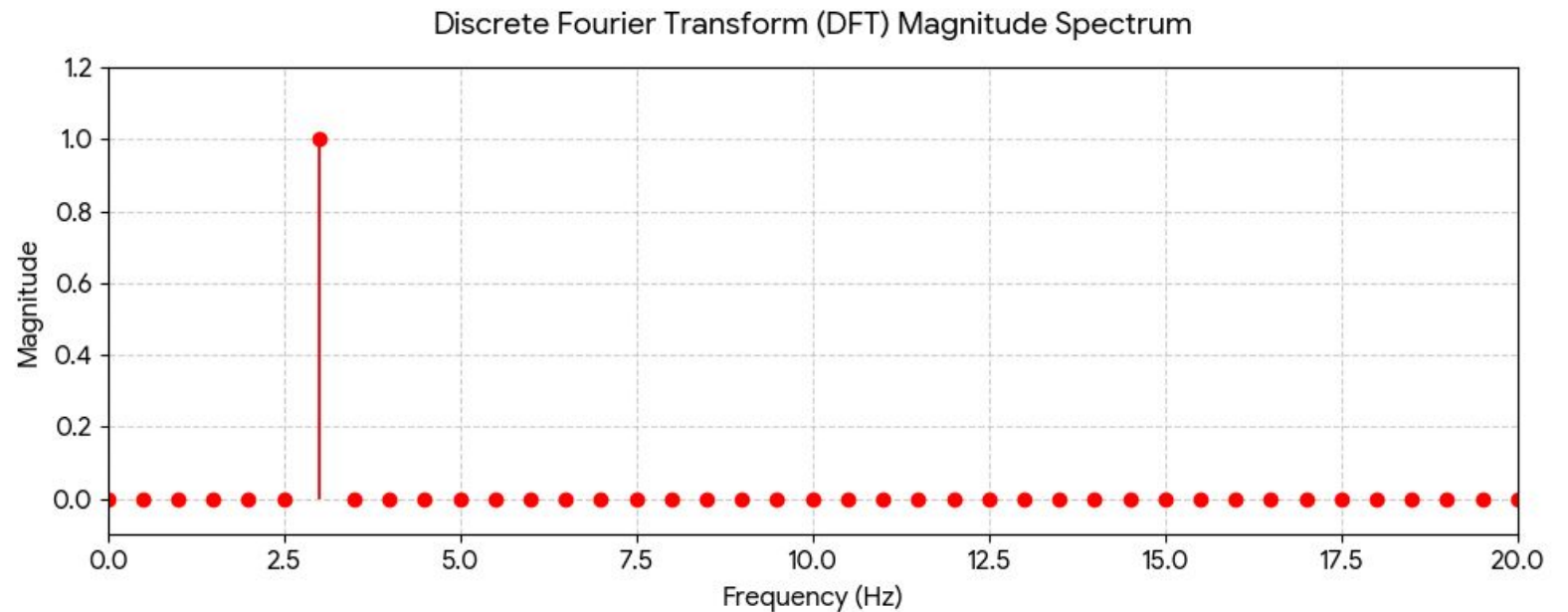
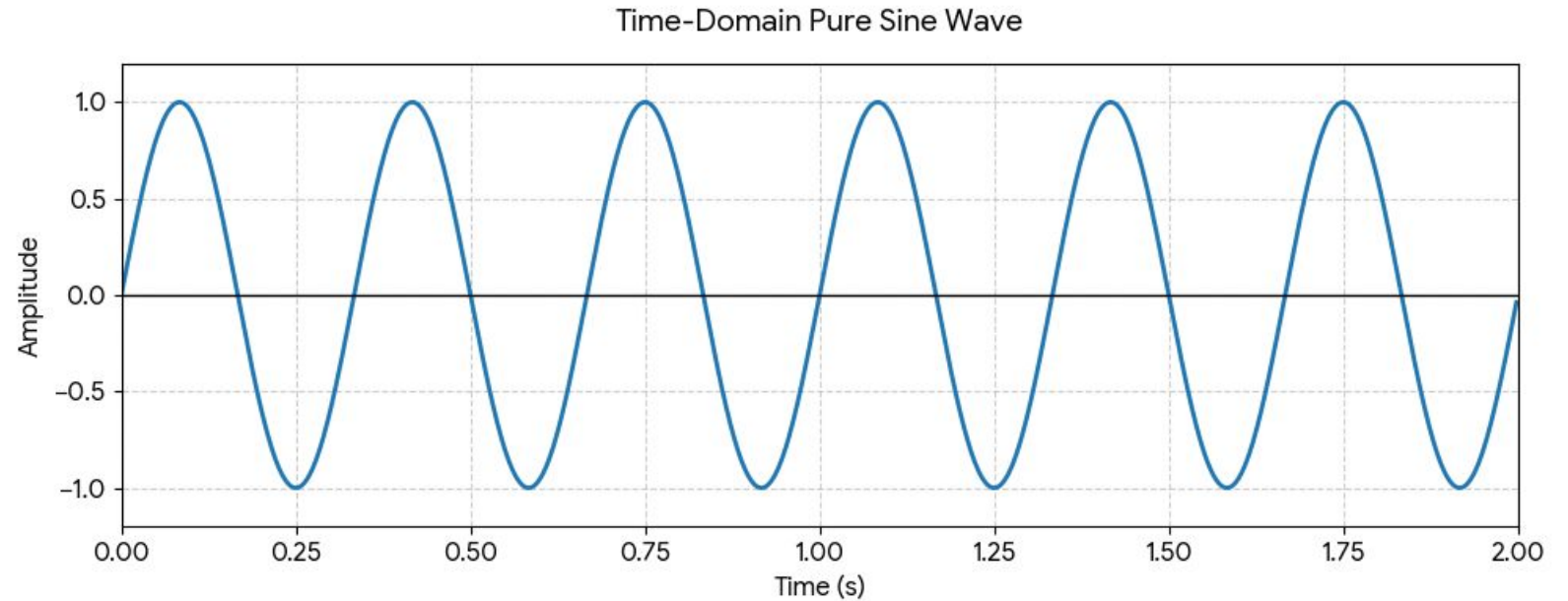
Feature Extraction

- Feature extraction is the process of converting a raw speech signal into a compact, robust sequence of numerical vectors.
- Raw speech waveforms are highly variable, contain immense noise, and have massive data rates.
- Feature extraction condenses the signal into a parametric representation with a minimized data rate.
- It discards irrelevant background noise and speaker variability while preserving the critical phonetic or acoustic information.

Feature Extraction

Example:

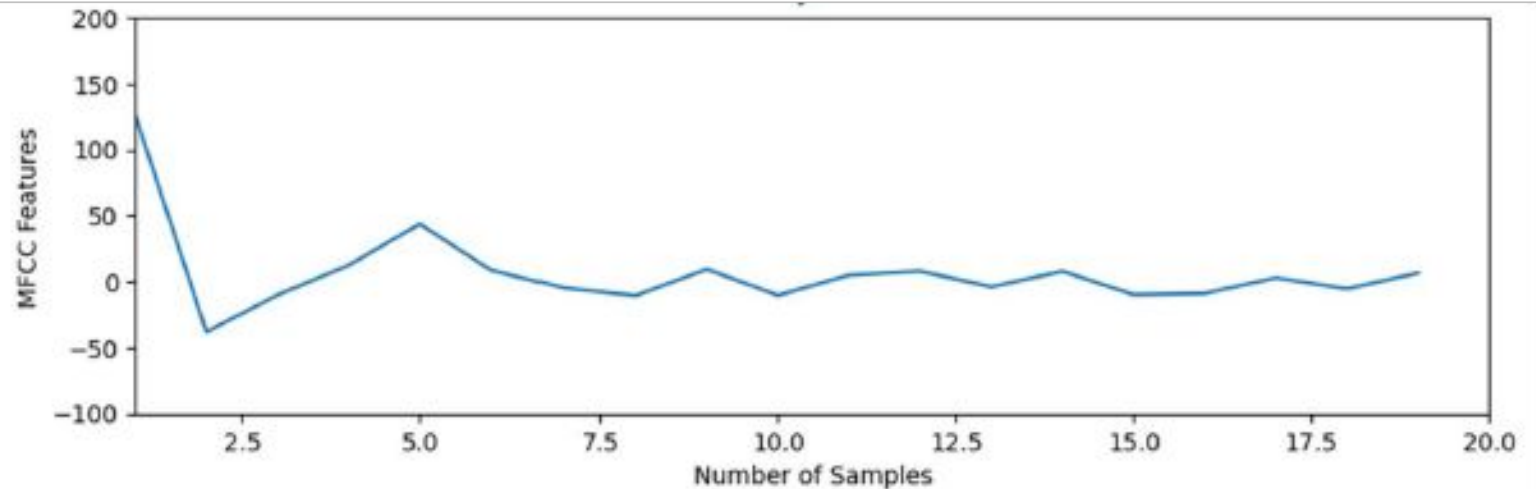
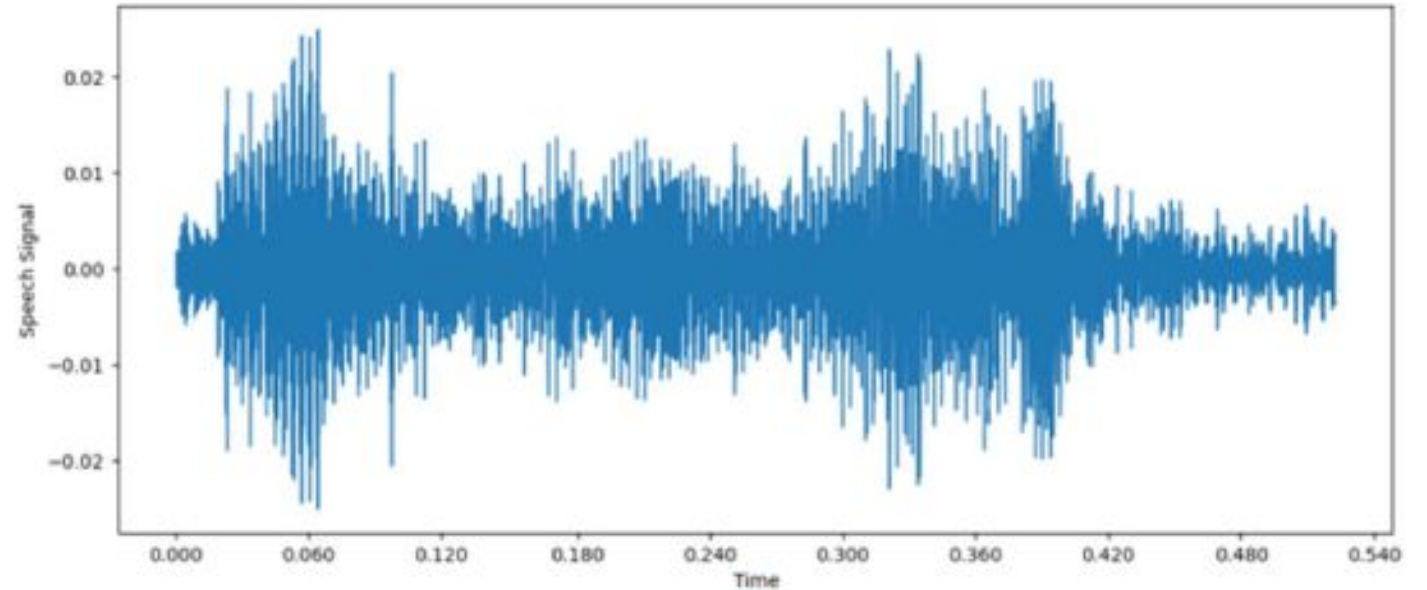
- Using DFT -
from a sine wave



Feature Extraction

Example:

- Using MFCC -
from a speech signal



Types of Features

- Features are of various types:

A. Static Features:

- It is characteristics extracted from a segment of data that describe its overall properties over a fixed window of time, ignoring how the signal changes moment to moment.

- Different types of static features are

1. Acoustic Features:

- Pitch, Energy, and Duration.

2. Prosodic / Long-Term Features:

- Rhythm, intonation, and stress patterns (utterance-level).

3. Temporal (Time-Domain) Features:

- Root Mean Square (RMS) Energy, Zero-Crossing Rate

Types of Features

- Features are of various types:
 - A. Static Features:
 - 4. Spectral (Frequency-Domain) Features:
 - Spectrogram, MFCCs
 - 5. Glottal Features
 - Jitter, Harmonics-to-Noise Ratio
 - 6. Deep Learning & Learned Embeddings:
 - Convolution Neural Network (CNN).

Types of Features

- Features are of various types:

B. Dynamic Features :

- Dynamic features capture how the acoustic properties of a voice change over time.
- Dynamic features track the movement, speed, and acceleration of the speech articulators (tongue, lips, jaw) across successive frames.
- First and second derivatives of features to track how the sound changes over time.
- Types:
 1. Delta:
 - * First-order temporal derivatives of the static features.
 - * Measure the rate of change of the spectrum from one frame to the next.
 2. Delta-Delta:
 - * Second-order temporal derivatives, calculated by taking the derivative of the Delta features.

Acoustic Features

Acoustic Features:

- Physical properties of the speech such as Pitch (frequency), Energy (loudness), and Duration.
- It is divided in to various types such as temporal, spectral, cepstral and prosodic features.

Temporal Features

Temporal (Time-Domain) Features:

- Metrics extracted directly from the raw waveform over time.
- Types:
 1. Root Mean Square (RMS) Energy: Measures amplitude/loudness variations across frames.
 2. Zero-Crossing Rate (ZCR): Tracks how often a signal changes sign (+ to -).
- It is used for distinguishing voiced speech (low ZCR, high energy) from unvoiced speech / silence (high ZCR, low energy).

Spectral Features

Spectral (Frequency-Domain) Features:

- Features derived by transforming the signal into the frequency domain using Short-Time Fourier Transform (STFT)
 - Spectrogram: A visual map plotting frequency over time, mapping signal intensity.
 - Spectral Centroid: Pinpoints the "center of mass" of the spectrum; correlates to sound "brightness".
 - Spectral Bandwidth: Measures the frequency spread around the centroid.

Cepstral Features

Cepstral Features:

- Features warped mathematically to mimic how the human ear actually filters and perceives sound scales.
 - MFCC:
 - * Mel-Frequency Cepstral Coefficients (MFCCs) are specialized features that mimic how human ears process frequencies that are used for speech recognition software.
 - LPCC:
 - * Vocal tract filter estimation.
 - * Models the vocal tract as a digital filter, representing the resonant frequencies (formants) of speech.

Prosodic Features

- Spans across multiple phone segments, syllables, or entire sentences.
- They carry emotional and stylistic context.
- Types:
 1. Fundamental Frequency (F_0 / Pitch): The rate of vocal cord vibration; reflects intonation and stress.
 2. Duration / Rhythm: Speaking rate, pause durations, and vowel elongation.
 3. Intensity: Variations in volume over long segments.
- Used in Speech Emotion Recognition (SER), emphasis detection, and speech synthesis (TTS).

Glottal Features

- Measure the state, tension, and structural health of the speaker's vocal folds (glottis).
- Jitter:
Short-term variations in the pitch (F_0) period (frequency instability).
- Shimmer:
Short-term variations in the peak-to-peak amplitude (loudness instability).
- Harmonics-to-Noise Ratio (HNR):
Ratio of periodic energy to acoustic noise.
- Used in Clinical speech pathology (detecting voice disorders), age/gender classification, and speaker identification.

Deep Learning Features

- Speech recognition uses deep learning features.

- Some of the features are:

1. Spectrogram Representations:

Deep neural networks (CNNs) treat 2D spectrograms as images.

2. Self-Supervised Embeddings:

Dense vector representations extracted from massive pre-trained models (e.g., wav2vec 2.0, HuBERT, Whisper embeddings).

3. Speaker Embeddings:

Vectors designed specifically to isolate speaker identity (e.g., d-vectors, x-vectors).